

Metals Comment: Battery metals bear market continues

European lithium industry will require more investment. We attended the Fastmarkets European Battery Raw Materials Conference this week in Amsterdam. The barriers facing European lithium project developments and the future of the European battery supply chain were key areas of focus, especially given US IRA incentives. Conference participants highlighted the key challenges in the form of unconventional European lithium deposits (mica and geothermal brines), long lead times, poor social acceptance of mining projects and increased competition to attract capital, alongside a lack of risk appetite for raw material projects in Europe. Whilst the Critical Raw Materials Act's (CRMA) new regulations, which lay the groundwork for faster permitting (12 months for processing/recycling and 24 months for extraction projects), was noted by many participants, alongside improved support from national governments, there was widespread frustration over a perceived lack of targeted policy support and funding access. It has appeared likely for some time that Europe cannot gain lithium supply independence this decade, given the lack of traditional deposits and lagging supply plans. However, some degree of long-term independence can still be achieved if actionable policy support is put into place for both extraction and processing, in our view. Finally, the EU investigation into Chinese EVs was top of mind, and participants highlighted the importance of China's consumer sector for EU exports as well as Chinese investment in Europe, which they believed meant higher tariffs would be unlikely. Our equity analysts have also highlighted in the past that high Chinese FDI in the EU battery supply chain may refrain EU policymakers from imposing heavy trade barriers.

Feedstock competition can weigh on refining plans. On our estimates, European lithium demand will reach ~550kt LCE by 2030, which implies 220kt of localised processed supply and 55kt of localised upstream supply under the CRMA. Currently, the total announced capacity of extraction projects in Europe stands at ~200kt LCE; however, we include just 10kt of upstream supply in our base case, with the probable capacity (~120kt LCE) contingent on funding access, permitting and social licenses to operate locally. On the processing side, presently, no asset is fully operational in Europe, with first capacity coming online at the end of this year, in our view. Our discussions at the conference highlighted several processing projects are in the pipeline, with most projects expected to start production after 2026E. Whilst we see a growing surplus in the lithium market over 2024-2028 due to growth mostly driven by China, Australia, Argentina and Zimbabwe, we note that feedstock competition is expected to remain intense, given the current agreements show upstream supply destined for Chinese, Korean and Japanese refineries. Therefore,

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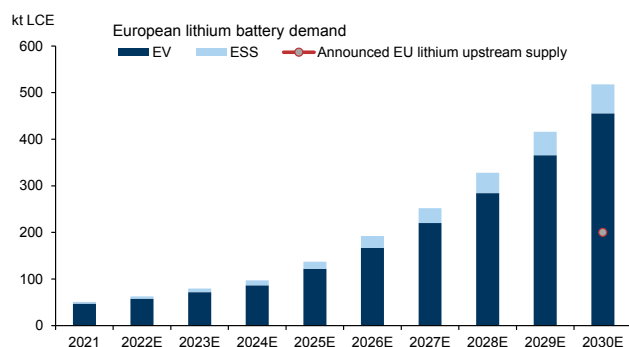
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we think that it could be advantageous for the EU to create an attractive landscape for the midstream supply chain to gain access to feedstock in order to serve OEM offtakes. Broadly speaking, the ex-China refining industry will likely need to become more flexible in terms of its ability to take various feedstock - technical grade carbonate and sulphate intermediates. Absent supply investments in both the upstream and midstream lithium supply chain, there is scope for the EU's plans for self-sufficiency to be delayed, in our view.

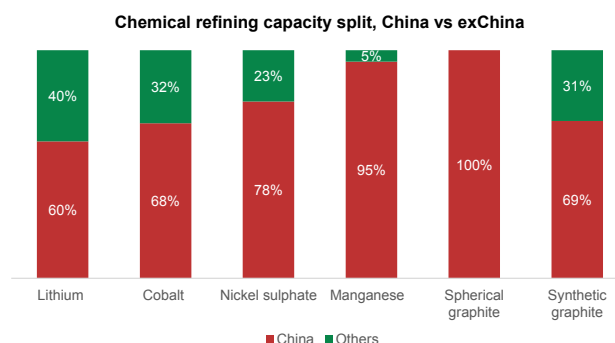
Battery metal prices expected to decline. Many industry participants see lithium prices continuing their decline, with 2024 seen as the year of oversupply. We note that the onshore China lithium carbonate price has reached 174k RMB/t (or \$21k/t excluding VAT), and the September lithium carbonate CIF CJK contract is now trading at \$27.5k/t. Going forward, we continue to forecast China lithium carbonate price to move lower to \$15k/LCE over the next 12 months and expect east Asian prices to follow the suite. For the rest of the year, the fundamental focus appears to be on a seasonal uptick in Chinese EV sales, and we think any disappointment relative to historical norms could put accelerated downward pressure on prices. Whilst there was broad agreement with our outlook for lithium carbonate prices, some participants noted the cost of production in China using lepidolite resources to be higher than our estimates. Finally, African projects were noted by some at the conference to be ramping up faster than market expectations. On a pre-disruption basis, we estimate 64kt of lithium supply coming out of Zimbabwe in 2023 which will rise to 180kt in 2024, representing 20% of global spodumene supply. On cobalt, most participants believed the market is oversupplied, in line with our view. Over the next few months, we expect cobalt metal prices to remain range bound, as we think risks of an upside surprise in demand remain low and supply from the DRC and Indonesia continue to rise. However, some participants noted that highly concentrated supply can lead to volatility in prices.

Exhibit 1: Whilst there is ~200kt of announced upstream capacity in Europe, our base case forecasts only 10kt of European upstream supply vs demand of 550kt by 2030E



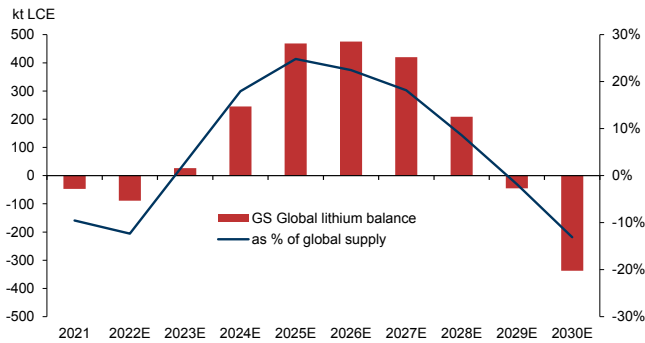
Source: Goldman Sachs Global Investment Research, Woodmac, BNEF, Company data

Exhibit 2: Upstream refining remains a bottleneck for Europe, but refining could also require a rise in investment to compete against China



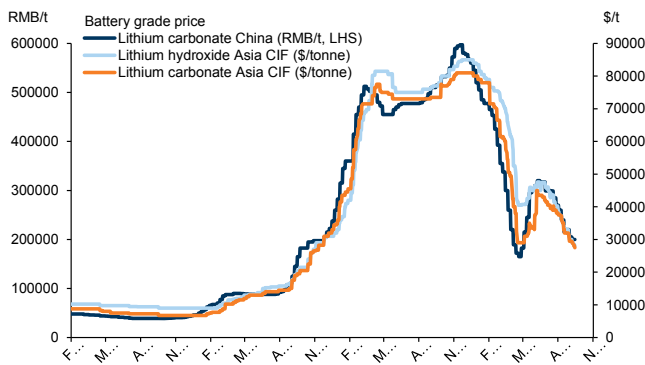
Source: Goldman Sachs Global Investment Research

Exhibit 3: We expect the lithium market to remain in surplus for the better part of this decade



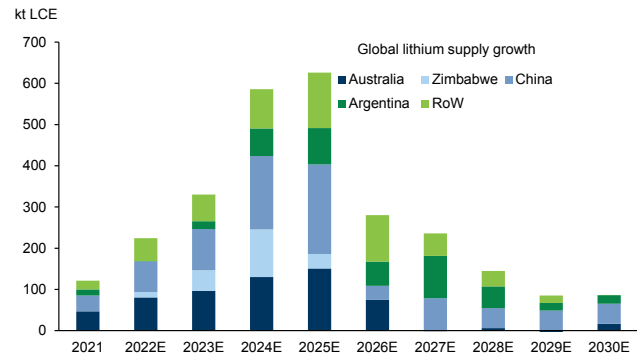
Source: Goldman Sachs Global Investment Research, Woodmac, SMM, Data compiled by Goldman Sachs Global Investment Research

Exhibit 5: Near-term, we expect prices to continue their downward trend on rising supply, onshore battery overcapacity and slowing demand growth



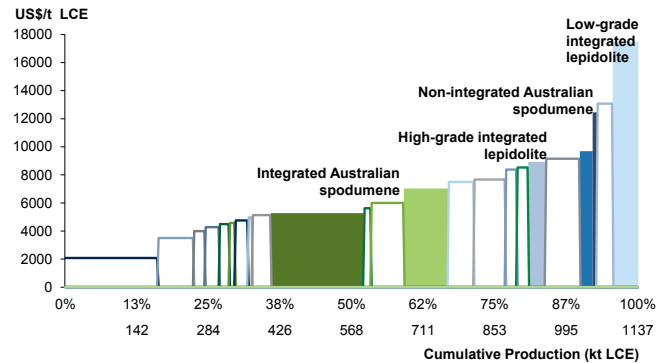
Source: Goldman Sachs Global Investment Research, SMM, Fastmarkets

Exhibit 4: Rising supply is tied to China both via domestic integrated projects and offshore joint ventures/offtakes



Source: Goldman Sachs Global Investment Research, Company data, BNEF

Exhibit 6: Lithium prices are currently trading over the industry cost curve, which is at odds with the supply rationing that would be needed to restrain the expected glut next year
2024 cost curve, excluding royalties and VAT



Source: Goldman Sachs Global Investment Research, Data compiled by Goldman Sachs Global Investment Research, Woodmac

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